

Ashish Kumar

Embedded C++ Engineer — Firmware Development & Bare-Metal Systems

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TECHNICAL SKILLS

- **Programming Languages:** C (C99/C11), Modern C++ (C++17), Assembly (ARM Cortex-M), Data Structures & Algorithms.
- **C++ Concepts:** OOP, STL, RAII, Smart Pointers, Move Semantics, Pointers, Memory Alignment, Cache Optimization.
- **Microcontrollers & Hardware:** STM32 (Cortex-M4), ESP32, Bare-Metal Driver Development, Register Manipulation, Interrupts, Timers.
- **Communication Protocols:** UART, SPI, I2C, CAN Bus, DMA (Direct Memory Access).
- **RTOS & Systems:** FreeRTOS (Tasks, Queues, Semaphores, Mutexes), Concurrency, Memory Management, Volatile Keyword.
- **Tools & Debugging:** GDB, OpenOCD, Logic Analyzers, Oscilloscopes, Git, Make, CMake, Linux (Arch / Ubuntu Terminal).

KEY SYSTEMS PROJECTS

Persistence-Only Disk Storage Engine

C++17, System Programming

- Built a high-performance disk serialization layer utilizing offset-based pointers instead of raw pointers for state persistence.
- Eliminated pointer-swizzling overhead during state recovery, reducing disk reload latency and optimizing memory footprint.
- Implemented manual memory management and explicit layout contiguity to minimize cache misses.

STM32 Bare-Metal Signal Generator & Peripherals

Embedded C, STM32, Hardware

- Developed register-level drivers for DAC, ADC, and DMA peripherals without relying on HAL libraries for maximum performance.
- Configured hardware timer interrupts for precise microsecond signal generation, verified using a logic analyzer.
- Optimized interrupt service routines (ISRs) to maintain minimal execution latency under system load.

EDUCATION

Bachelor of Technology (B.Tech) / Relevant Degree

India

Focus on Embedded Systems, Computer Architecture, and Low-Level Programming